

Pattern Recognition Techniques for Environmental Sciences

Using the CUED template



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I would like to dedicate this thesis to my loving parents ...

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This dissertation contains fewer than 65,000 words including appendices, bibliography, footnotes, tables and equations and has fewer than 150 figures.

Niclas Rieger
December 2024

Acknowledgements

And I would like to acknowledge ...

Abstract

This is where you write your abstract ...

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Part I

hello

Chapter 1

Introduction

Lorem Ipsum is simply dummy [\[1\]](#)

$$E = mc^2 \int_{x=2}^{x=5} x^2 dx \quad (1.1)$$

It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using 'Content here, content here', making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for 'lorem ipsum' will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).

1.1 Where does it come from?

Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, *consectetur*, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of "de Finibus Bonorum et Malorum" (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, "Lorem ipsum dolor sit amet..", comes from a line in section 1.10.32.

The standard chunk of Lorem Ipsum used since the 1500s is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from "de Finibus Bonorum et Malorum" by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham

"Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris

nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum."

Chapter 2

Theoretical Foundations of Pattern Recognition Techniques

2.1 Principal Component Analysis

2.2 Canonical Correlation Analysis

2.3 Gaussian Processes

2.4 Integrative Perspective

Part II

Hello 2

Chapter 3

Application to Environmental Problems

3.1 Shared Dynamics between Temporally Lagged Climate Variables: A Case Study of SST and Continental Precipitation

References

1. Stegmann, P., Daioglou, V., Londo, M., Van Vuuren, D. P. & Junginger, M. Plastic futures and their CO₂ emissions. *Nature* **612**, 272–276. doi:[10.1038/s41586-022-05422-5](https://doi.org/10.1038/s41586-022-05422-5) (2022) (cit. on p. 3).

Appendix A

How to install L^AT_EX

Windows OS

TeXLive package - full version

Appendix B

Installing the CUED class file

$\TeX$.cls files can be accessed system-wide when they are placed in the `<texmf>/tex/latex` directory, where `<texmf>` is the root directory of the user's $\TeX$ installation. On systems that have a local texmf tree (`<texmflocal>`), which may be named “texmf-local” or “localtexmf”, it may be advisable to install packages in `<texmflocal>`, rather than `<texmf>` as the contents of the former, unlike that of the latter, are preserved after the $\TeX$ system is reinstalled and/or upgraded.

It is recommended that the user create a subdirectory `<texmf>/tex/latex/CUED` for all CUED related $\TeX$ class and package files. On some $\TeX$ systems, the directory look-up tables will need to be refreshed after making additions or deletions to the system files. For $\TeX$ Live systems this is accomplished via executing “texhash” as root. $\text{MiK}\TeX$ users can run “initexmf -u” to accomplish the same thing.

Users not willing or able to install the files system-wide can install them in their personal directories, but will then have to provide the path (full or relative) in addition to the filename when referring to them in $\TeX$.

